COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9-17-26	**DATE SUBMITTED:**	9-17-26
NAME:	Masongsong, John Marvin B	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 8.1 **Install, Configure, and Manage Enterprise Availability Monitoring via Ansible**

1. PROCEDURE OUTPUT

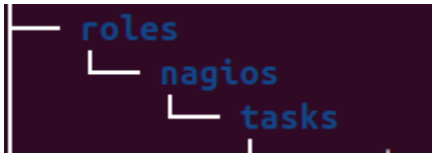
1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.

2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

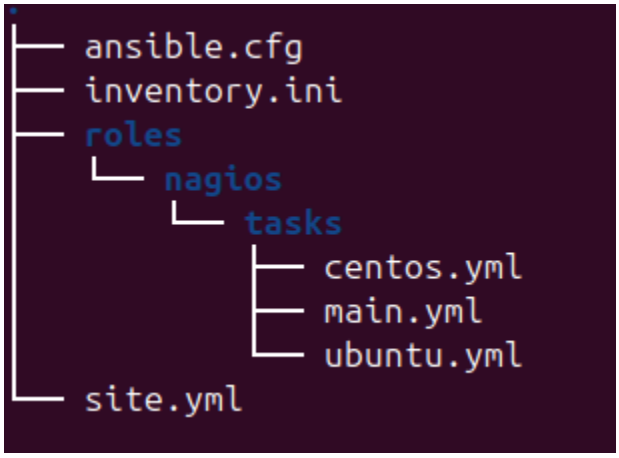
3. Show an output of the installed Nagios for both Ubuntu and CentOS.

4. Make sure to create a new repository in GitHub for this activity.

Create the Ansible Role



Create the Task Files



Configure the Inventory

```
[ubuntu]
192.168.56.104 ansible_user=masongsong

[centos]
192.168.56.102 ansible_user=masong
```

Configure ansible.cfg

```
[defaults]
inventory = inventory.ini

interpreter_python = auto_silent

deprecation_warnings = False
```

Create the Main Task File

```
---
- name: Include Ubuntu tasks
  include_tasks: ubuntu.yml
  when: ansible_os_family == "Debian"

- name: Include CentOS tasks
  include_tasks: centos.yml
  when: ansible_os_family == "RedHat"
```

Install Nagios Dependencies

Ubuntu:

```
- name: Install Nagios dependencies
  ansible.builtin.apt:
    name:
      - autoconf
      - gcc
      - libc6
      - make
      - wget
      - unzip
      - apache2
      - php
      - libapache2-mod-php
      - libgd-dev
      - ufw
      - openssl
      - libssl-dev
    state: present
```

CentOS:

```
---
- name: Install Nagios dependencies
  ansible.builtin.dnf:
    name:
      - autoconf
      - gcc
      - glibc
      - glibc-common
      - make
      - wget
      - unzip
      - httpd
      - php
      - gd
      - gd-devel
      - openssl
      - openssl-devel
      - tar
    state: present
```

Create Nagios Group

Ubuntu:

```
- name: Create Nagios group
  ansible.builtin.group:
    name: nagios
    state: present
```

CentOS:

```
- name: Create Nagios group
  ansible.builtin.group:
    name: nagios
    state: present
```

Create Nagios User

Ubuntu:

```
- name: Create Nagios user
  ansible.builtin.user:
    name: nagios
    group: nagios
    state: present
```

CentOS:

```
- name: Create Nagios user
  ansible.builtin.user:
    name: nagios
    group: nagios
    state: present
```

Add www-data to nagios group

Ubuntu:

```
- name: Add www-data to Nagios group
  ansible.builtin.user:
    name: www-data
    groups: nagios
    append: yes
```

CentOS: (Add apache to nagios group)

```
- name: Add apache to Nagios group
  ansible.builtin.user:
    name: apache
    groups: nagios
    append: yes
```

Download Nagios Core

Ubuntu:

```
- name: Download Nagios Core
  ansible.builtin.get_url:
    url: https://github.com/NagiosEnterprises/nagioscore/archive/refs/tags/nagi>
    dest: /tmp/nagioscore.tar.gz
```

CentOS:

```
- name: Download Nagios Core
  ansible.builtin.get_url:
    url: https://github.com/NagiosEnterprises/nagioscore/archive/refs/tags/nagi>
    dest: /tmp/nagioscore.tar.gz
```

Extract Nagios Core

Ubuntu:

```
- name: Extract Nagios Core
  ansible.builtin.unarchive:
    src: /tmp/nagioscore.tar.gz
    dest: /tmp
    remote_src: yes
```

CentOS:

```
- name: Extract Nagios Core
  ansible.builtin.unarchive:
    src: /tmp/nagioscore.tar.gz
    dest: /tmp
    remote_src: yes
```

Find Nagios Source Directory

Ubuntu:

```
- name: Find Nagios source directory
  ansible.builtin.shell: "find /tmp -maxdepth 1 -type d -name 'nagioscore-nagio
  register: nagios_dir
  changed_when: false
```

CentOS:

```
- name: Find Nagios source directory
  ansible.builtin.shell: "find /tmp -maxdepth 1 -type d -name 'nagioscore-nagio
  register: nagios_dir
  changed_when: false
```

Configure Nagios

Ubuntu:

```
- name: Configure Nagios
  ansible.builtin.command:
    cmd: ./configure --with-httpd-conf=/etc/apache2/conf-available
    chdir: "{{ nagios_dir.stdout }}"
```

CentOS:

```
- name: Configure Nagios
  ansible.builtin.command:
    cmd: ./configure --with-httpd-conf=/etc/httpd/conf.d
    chdir: "{{ nagios_dir.stdout }}"
```

Compile Nagios

Ubuntu:

```
- name: Compile Nagios
  ansible.builtin.command:
    cmd: make all
    chdir: "{{ nagios_dir.stdout }}"
```

CentOS:

```
- name: Compile Nagios
  ansible.builtin.command:
    cmd: make all
    chdir: "{{ nagios_dir.stdout }}"
```

Install Nagios

Ubuntu:

```
- name: Install Nagios
  ansible.builtin.command:
    cmd: make install
    chdir: "{{ nagios_dir.stdout }}"
```

CentOS:

```
- name: Install Nagios
  ansible.builtin.command:
    cmd: make install
    chdir: "{{ nagios_dir.stdout }}"
```

Install Nagios Command Mode

Ubuntu:

```
- name: Install Nagios command mode
  ansible.builtin.command:
    cmd: make install-commandmode
    chdir: "{{ nagios_dir.stdout }}"
```

CentOS:

```
- name: Install Nagios command mode
  ansible.builtin.command:
    cmd: make install-commandmode
    chdir: "{{ nagios_dir.stdout }}"
```

Install Nagios Configuration

Ubuntu:

```
- name: Install Nagios configuration
  ansible.builtin.command:
    cmd: make install-config
    chdir: "{{ nagios_dir.stdout }}"
```

CentOS:

```
- name: Install Nagios configuration
  ansible.builtin.command:
    cmd: make install-config
    chdir: "{{ nagios_dir.stdout }}"
```

Install Nagios Web Configuration

Ubuntu:

```
- name: Install Nagios web configuration
  ansible.builtin.command:
    cmd: make install-webconf
    chdir: "{{ nagios_dir.stdout }}"
```

CentOS:

```
- name: Install Nagios web configuration
  ansible.builtin.command:
    cmd: make install-webconf
    chdir: "{{ nagios_dir.stdout }}"
```

Create Nagios web authentication file

Ubuntu:

```
- name: Create Nagios web authentication file
  ansible.builtin.command:
    cmd: htpasswd -bc /usr/local/nagios/etc/htpasswd.users nagiosadmin Prowitht>
    creates: /usr/local/nagios/etc/htpasswd.users
```

CentOS:

```
- name: Create Nagios web authentication file
  ansible.builtin.command:
    cmd: htpasswd -bc /usr/local/nagios/etc/htpasswd.users nagiosadmin Prowitht>
    creates: /usr/local/nagios/etc/htpasswd.users
```

Enable Apache Modules

Ubuntu:

```
- name: Enable Apache rewrite module
  ansible.builtin.command:
    cmd: a2enmod rewrite

- name: Enable Apache CGI module
  ansible.builtin.command:
    cmd: a2enmod cgi
```

CentOS:

Configure Firewall

Ubuntu:

```
- name: Allow Apache through UFW
  ansible.builtin.command:
    cmd: ufw allow Apache
```

CentOS:

```
- name: Configure firewall
  ansible.builtin.command:
    cmd: firewall-cmd --permanent --add-service=http
```

Reload UFW

Ubuntu:

```
- name: Reload UFW
  ansible.builtin.command:
    cmd: ufw reload
```

CentOS:

```
- name: Reload firewalld
  ansible.builtin.command:
    cmd: firewall-cmd --reload
```

Install Nagios Daemon

Ubuntu:

```
- name: Install Nagios daemon
  ansible.builtin.command:
    cmd: make install-daemoninit
    chdir: "{{ nagios_dir.stdout }}"
```

CentOS:

```
- name: Install Nagios daemon
  ansible.builtin.command:
    cmd: make install-daemoninit
    chdir: "{{ nagios_dir.stdout }}"
```

Start Apache

Centos:

```
- name: Start Apache
  ansible.builtin.systemd_service:
    name: httpd
    state: started
    enabled: yes
```

Restart Apache

Ubuntu:

```
- name: Restart Apache
  ansible.builtin.systemd_service:
    name: apache2
    state: restarted
```

CentOS:

```
- name: Restart Apache
  ansible.builtin.systemd_service:
    name: httpd
    state: restarted
```

Start Nagios

Ubuntu:

```
- name: Start Nagios
  ansible.builtin.systemd_service:
    name: nagios
    state: started
```

CentOS:

```
- name: Start Nagios
  ansible.builtin.systemd_service:
    name: nagios
    state: started
    enabled: yes
```

Enable Nagios at Boot

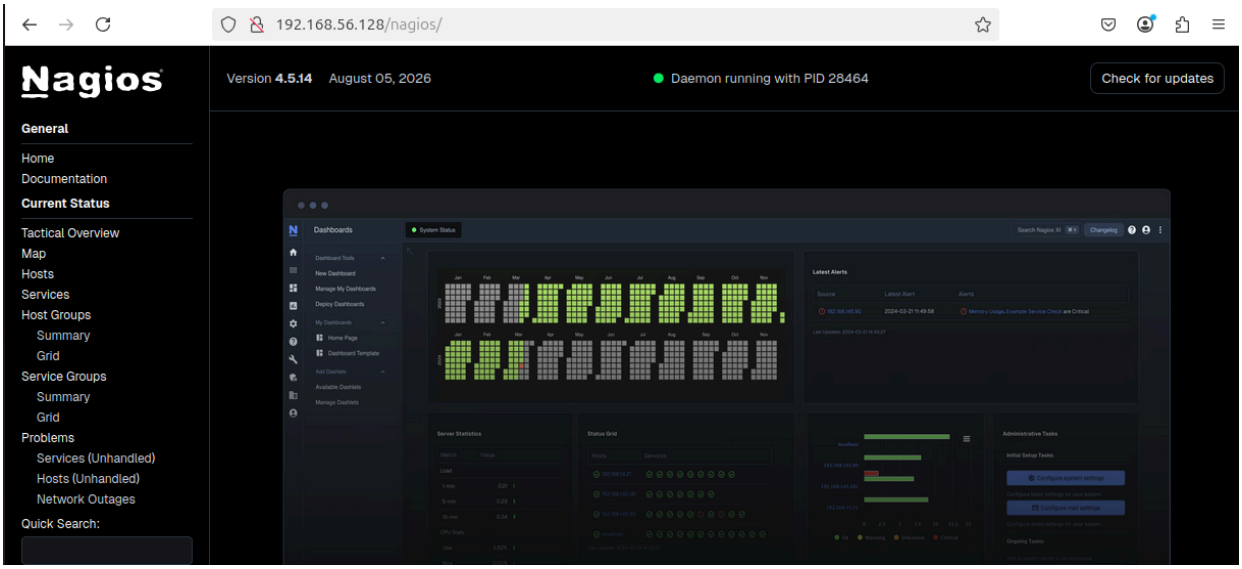
Ubuntu:

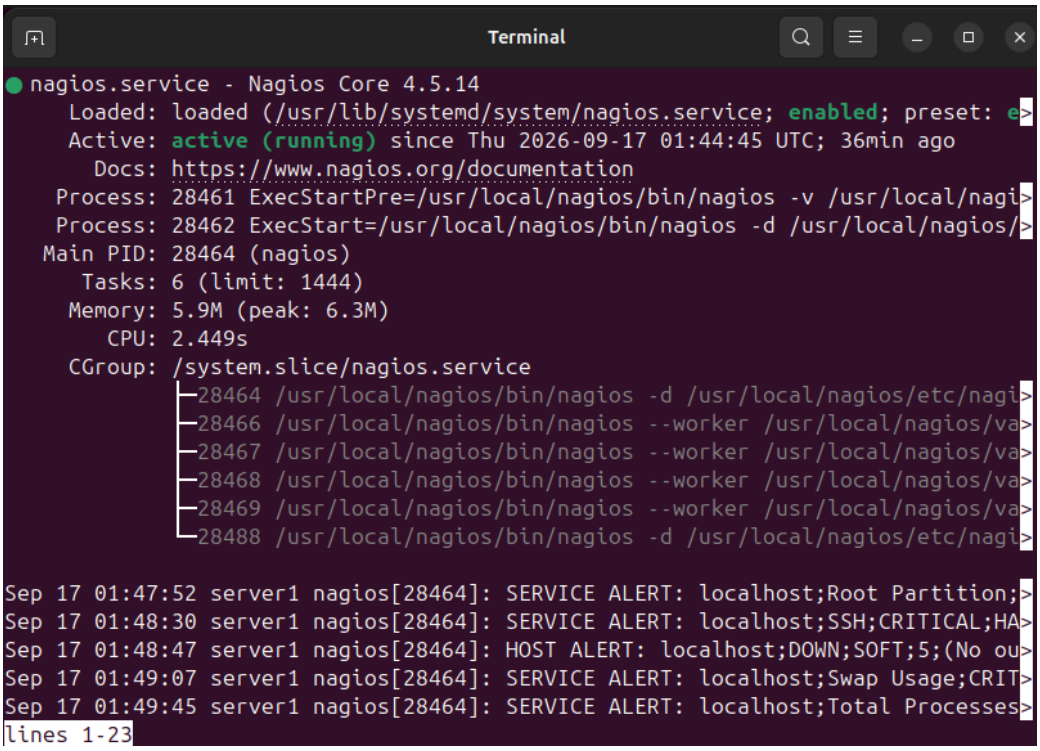
```
- name: Enable Nagios at boot
  ansible.builtin.systemd_service:
    name: nagios
    enabled: yes
```

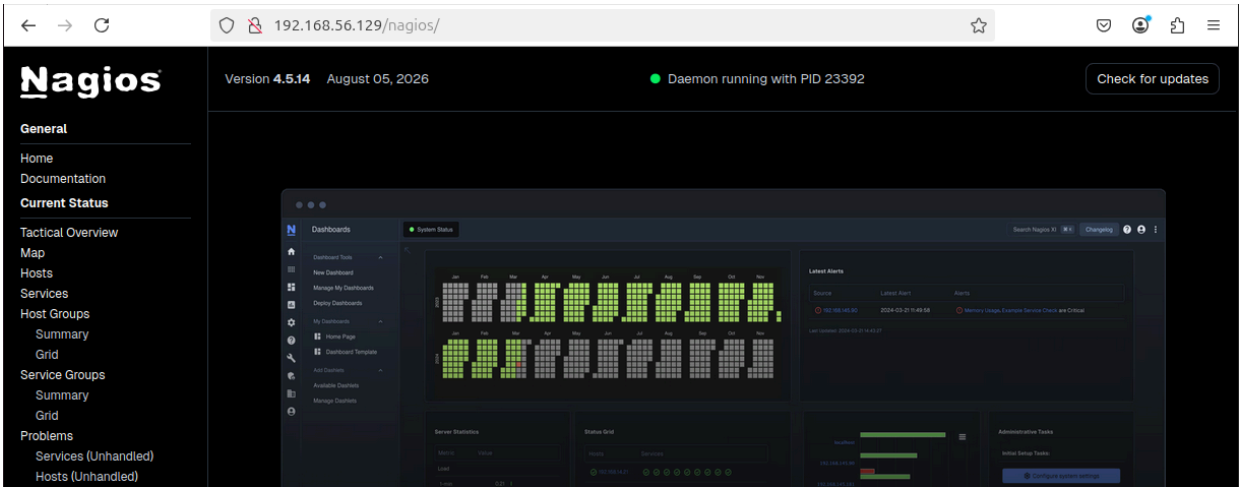
Ubuntu:

```
Terminal
● nagios.service - Nagios Core 4.5.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-09-17 01:44:45 UTC; 36min ago
       Docs: https://www.nagios.org/documentation
    Process: 23389 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/>
    Process: 23390 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nag>
   Main PID: 23392 (nagios)
      Tasks: 6 (limit: 1444)
     Memory: 5.7M (peak: 6.4M)
        CPU: 1.540s
     CGroup: /system.slice/nagios.service
             ├─23392 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             ├─23393 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/na>
             ├─23395 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/na>
             ├─23396 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/na>
             ├─23397 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/na>
             └─23421 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

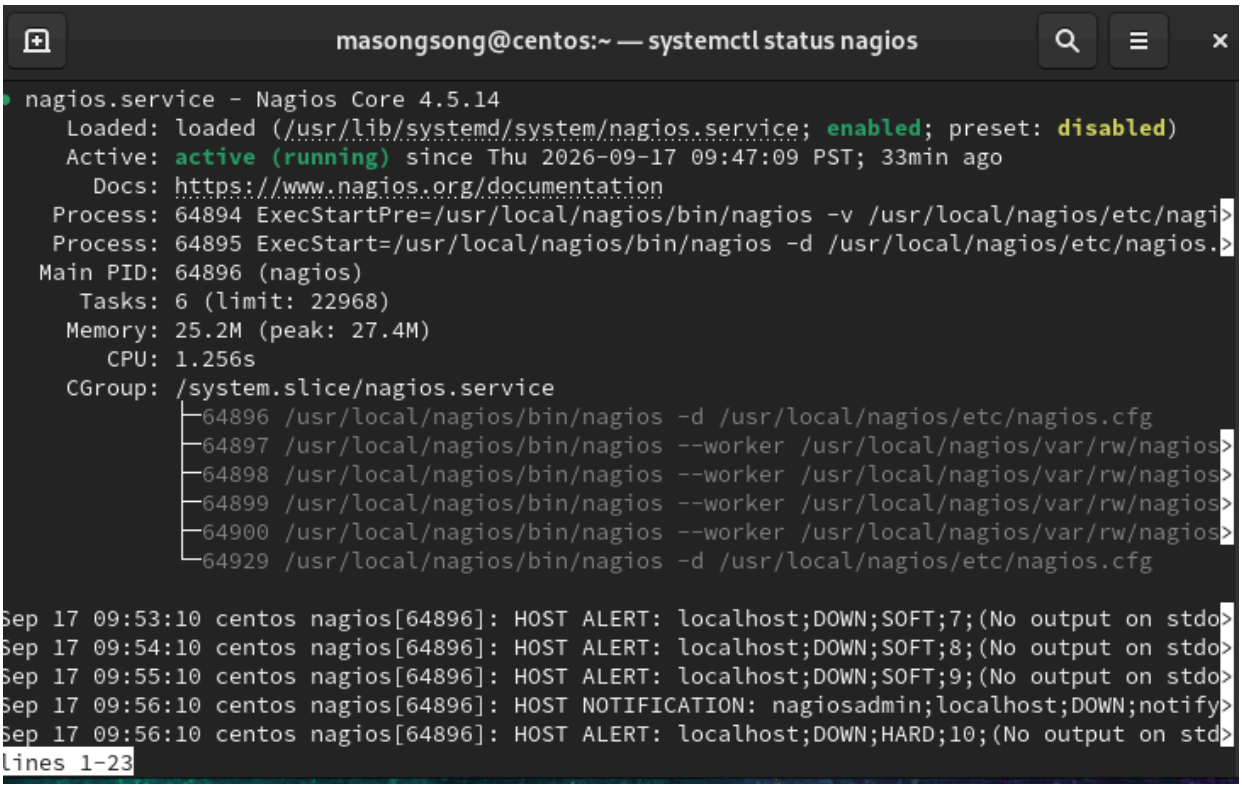
Sep 17 01:47:52 server2 nagios[23392]: SERVICE ALERT: localhost;Root Partition;CRITICA>
Sep 17 01:48:30 server2 nagios[23392]: SERVICE ALERT: localhost;SSH;CRITICAL;HARD;1;(N>
Sep 17 01:48:46 server2 nagios[23392]: HOST ALERT: localhost;DOWN;SOFT;5;(No output on>
Sep 17 01:49:07 server2 nagios[23392]: SERVICE ALERT: localhost;Swap Usage;CRITICAL;HA>
Sep 17 01:49:45 server2 nagios[23392]: SERVICE ALERT: localhost;Total Processes;CRITIC>
lines 1-23
```

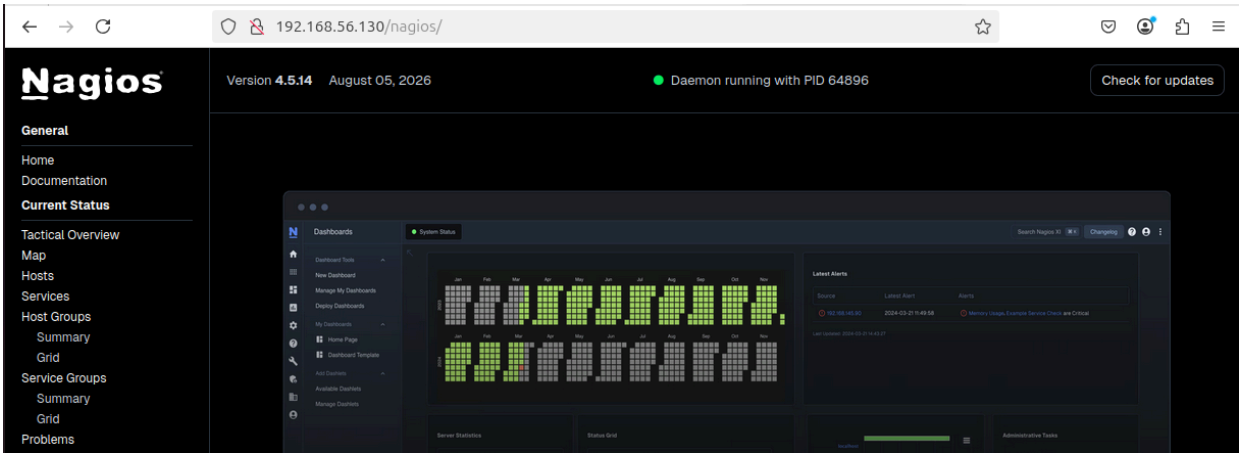

192.168.56.128/nagios/
Nagios
Version 4.5.14 August 05, 2026
Daemon running with PID 28464
Check for updates
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:


Terminal
nagios.service - Nagios Core 4.5.14
Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: e>
Active: active (running) since Thu 2026-09-17 01:44:45 UTC; 36min ago
Docs: https://www.nagios.org/documentation
Process: 28461 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagi>
Process: 28462 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/>
Main PID: 28464 (nagios)
Tasks: 6 (limit: 1444)
Memory: 5.9M (peak: 6.3M)
CPU: 2.449s
CGroup: /system.slice/nagios.service
├─28464 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi>
├─28466 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
├─28467 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
├─28468 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
├─28469 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/va>
└─28488 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagi>
Sep 17 01:47:52 server1 nagios[28464]: SERVICE ALERT: localhost;Root Partition;>
Sep 17 01:48:30 server1 nagios[28464]: SERVICE ALERT: localhost;SSH;CRITICAL;HA>
Sep 17 01:48:47 server1 nagios[28464]: HOST ALERT: localhost;DOWN;SOFT;5;(No ou>
Sep 17 01:49:07 server1 nagios[28464]: SERVICE ALERT: localhost;Swap Usage;CRIT>
Sep 17 01:49:45 server1 nagios[28464]: SERVICE ALERT: localhost;Total Processes>
lines 1-23


192.168.56.129/nagios/
Nagios
Version 4.5.14 August 05, 2026
Daemon running with PID 23392
Check for updates
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)

CentOS:


masongsong@centos:~ — systemctl status nagios
nagios.service - Nagios Core 4.5.14
Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
Active: active (running) since Thu 2026-09-17 09:47:09 PST; 33min ago
Docs: https://www.nagios.org/documentation
Process: 64894 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagi>
Process: 64895 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.>
Main PID: 64896 (nagios)
Tasks: 6 (limit: 22968)
Memory: 25.2M (peak: 27.4M)
CPU: 1.256s
CGroup: /system.slice/nagios.service
├─64896 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
├─64897 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios>
├─64898 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios>
├─64899 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios>
├─64900 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios>
└─64929 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
Sep 17 09:53:10 centos nagios[64896]: HOST ALERT: localhost;DOWN;SOFT;7;(No output on stdo>
Sep 17 09:54:10 centos nagios[64896]: HOST ALERT: localhost;DOWN;SOFT;8;(No output on stdo>
Sep 17 09:55:10 centos nagios[64896]: HOST ALERT: localhost;DOWN;SOFT;9;(No output on stdo>
Sep 17 09:56:10 centos nagios[64896]: HOST NOTIFICATION: nagiosadmin;localhost;DOWN;notify>
Sep 17 09:56:10 centos nagios[64896]: HOST ALERT: localhost;DOWN;HARD;10;(No output on std>
lines 1-23



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?

It is easier to just login using credentials and to see everything in just a website, rather than opening every single package just to see if they are up and running.

Conclusions/Learnings:

I have learned a lot from this HOA, mainly how to automate the installation and setup of nagios to both Debian and RedHat based Operating Systems. The use of my past seatwork about how to install nagios with commands has truly helped a lot into finishing this activity. I have mostly used ansible.builtin.” “ type commands in order to automate the needed tasks like installing and enabling dependencies/tools, extracting and downloading tar, creating user and groups, and much more. Honestly if I were to master this before I graduate, this would be a lot of help into bettering my chances in the future.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Asked for conversion of commands to ansible.builtin.” “ type command**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **The AI was used to convert the ad hoc commands to lines that are suitable for ansible.builtin.” “ type format of code.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

COMPUTER ENGINEERING

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